

THE GREAT STEM MYSTERIES

WHY TEN?

Mystery #2 — the number that has been following you



Art: A Saturday-morning bedroom quietly crowded with hidden tens — ten candles on a leftover slice of birthday cake, a TOP 10 poster, a test marked 10/10, two hands frozen mid high-five. Nothing looks suspicious. That's the point.

Here is a strange fact: you are being followed.

Not by a person. By a number.

Nia noticed it on an ordinary Saturday. Her cousin's leftover birthday cake had held ten candles. Her maths test said 10/10 — a good day. Her playlist was a Top Ten. Her fingers, when she counted them slowly, as if they might have changed overnight, came to ten.

They always came to ten.

"Ten out of ten," she said to the empty room. "Obviously."

Out in the world it was worse. Shops counted money in tens. Scientists measured in tens. Whole centuries were bundled into tens and given a special name — *decades* — as if that were the most natural thing in the universe.

Nia had lived with ten her entire life without once suspecting it of anything.

Then the clock on the wall ticked.

And the clock — the round-faced, innocent-looking clock she had trusted since she was five — was hiding something.

Art: Nia and her best friend Sam under a huge wall clock. Behind them, faint as ghosts: a Babylonian astronomer holding a clay tablet, a Maya scribe painting bars and dots, a modern computer blinking a single cold eye. All of them are looking at Nia's ten fingers.

Watch any clock. Count along with the seconds.

They don't gather in tens. They gather in *sixties*. Sixty seconds to a minute. Sixty minutes to an hour. Hiding in plain sight, on every wall on Earth.

"Sixty?" said Sam, who never believed anything without proof. "Who counts all the way to sixty before starting over?"

The Babylonians did. Four thousand years ago they counted in *sixties* — and they invented the minute you are living inside right now.

The Maya counted in *twenties*. Every computer on the planet counts in *twos* — and runs the world that way.

So other counting worlds exist. They built empires. They landed on the Moon. Which leaves an uncomfortable question, and Sam asked it with a pen in his hand:

Is ten actually special? YES or NO.

"Sign it," he said, drawing a box. "Ink means you mean it."

Sign yours too. Right here in the margin. We'll wait.

Art: A road forking at a battered signpost. One arm reads TEN IS SPECIAL. The other reads TEN JUST HAPPENED TO US. The second arm looks older, as if it has been waiting longer.

Nia signed YES. Almost everyone signs YES.

Then she looked at her signature for a while and asked the question that ruins afternoons.

"If the whole world forgot numbers tonight," she said, "and started fresh tomorrow morning... would ten win again?"

Think about that. Either ten earned its place — it's genuinely the best, and any smart civilisation would pick it again — or ten got lucky. Or something chose it that nobody talks about.

Say the question out loud to someone. Watch their face. First they'll shrug. Then they'll open their mouth to answer.

Then they'll stop.

"There's only one way to find out," said Sam, tearing a fresh page from his notebook. "We leave ten behind. Completely."

Art: A notebook page becoming a world. Pentaria: five-petalled flowers, five-sided houses, cheerful market crowds of one-handed Pentarians. Market bags everywhere, each holding exactly five.

Sam ruled a thick line across the page. "New world. New rule. Ten is banned."

"Banned?"

"Never existed. Everyone in this world has one hand. Five fingers. That's all they've ever counted on."

Nia drew the first citizen — one hand, big grin — and named the place Pentaria.

Pentarians count the way you learned to count in *The Mystery of Ten*: they bag things up. But their bags hold **five**. Five beans fill a bag. Bags first, loose ones after.

Try it — you're a Pentarian now. Take thirteen beans (or buttons, or anything). Bag them in fives. What do you get?

Two full bags. Three loose.

A Pentarian writes that as **23**. Two bags, three loose. Their "23" is our thirteen — and it isn't a mistake. It's their whole world, working.

"Fine. Counting works," said Sam, unimpressed. "Counting is the baby step. Can Pentaria do *maths*?"

Art: A Pentarian fruit market, mid-trade. A fifth bean is dropping toward four others — and a bag is snapping shut around them like a small satisfied mouth.

At a Pentarian market stall, a trader has four loose berries. A customer hands over three more.

Four loose plus three loose. Seven loose berries — except no Pentarian would ever say that.

"Watch," whispered Nia. "The bag —"

Five of the berries snap into a bag. Two stay loose. One bag, two loose: the trader writes **12**.

Their "12". Our seven. The carry — the exact trick you use when $8 + 5$ spills past nine — happened by itself, in a world where ten was never invented.

Your turn at the stall. In Pentarian: what is 3 loose + 4 loose? What about 14 + 3? (Answers in the back — plus the places your bags most likely leaked.)

Sam checked every answer against ordinary maths, certain something would snap.

It matched.

He tried bigger numbers. It matched. Subtraction. It matched.

It kept on matching.

Art: The signed YES/NO box from page 2, cracking across the middle like an eggshell. Nia and Sam stare at it. Nobody is touching it.

Sam put down his pencil the way a chess player puts down a lost king.

"Nothing broke," he said. "Nothing was ever going to break."

Say it plainly, because it deserves plain words: **mathematics does not care about ten.** Bags of five work. The Maya's bags of twenty work. Babylon's bags of sixty work. Adding, carrying, subtracting — the machine runs perfectly on any size of bag.

And nobody proved this to you. You did it yourself, at a market stall in a world you invented.

Now look back at page 2, at the box you signed in ink.

Keep your signature, or cross it out. Your choice. Sam crossed his out — slowly, like an apology.

"But hang on," said Nia, and her voice had gone quiet. "If mathematics didn't choose ten..."

She looked at the clock. She looked at her hands.

"...then something else did."

Art: The bedroom gone noir. Desk lamp like a spotlight. A world map spread out, countries drawn in numerals, and a magnifying glass sliding across it. Nia in a detective's pose she will deny ever striking.

Here is what makes it strange.

Egypt landed on ten. So did China. So did India, and Rome, and civilisations separated by oceans and centuries — people who never met, never traded, never shared a single word.

Almost all of them, independently, bagging the world in tens.

"That's not luck," said Nia. "Empires that never met don't all pick the same number out of a hat. That's a *clue*."

When mathematics itself doesn't force an answer, but the whole world agrees on one anyway, the reason is hiding somewhere outside mathematics. Somewhere so close to every human who ever counted that nobody thought to look at it.

Sam pulled the lamp lower, and the two of them did what every good detective does when the mystery stops making sense.

"Suspects," he said. "We write down suspects."

Art: A corkboard by lamplight. Suspect cards pinned in rows: TEN IS THE ROUNDEST NUMBER — SOME ANCIENT KING DECREED IT — EVERYONE COPIED ONE VILLAGE — and one card, slightly crooked: ...FINGERS? One card is blank, waiting.

They pinned up every suspect they could think of.

Suspect one: ten is the roundest, friendliest number, and people can feel it. (Nia liked this one.)

Suspect two: some ancient king decreed it, and it stuck.

Suspect three: one clever village invented ten and everyone else copied.

Suspect four — Sam pinned it slightly crooked, like he wasn't sure it deserved the board — *fingers?*

The last card is blank. It's yours.

Write your suspect. In ink. Ink means you mean it — and a detective who won't commit to a guess never learns anything, because a guess that can't be wrong can't be right either.

"Now what?" said Nia.

Sam tapped the board. "Now we remember the rule. A suspect is only a suspect until it *predicts* something."

Art: The corkboard again — but each suspect card has sprouted an arrow leading to a prediction card. The FINGERS arrow glows faintly, pointing at a drawing of hands across a world map.

A guess that predicts nothing is only a decoration.

So they made each suspect stick its neck out. *If you're the culprit, what should the world look like?*

If ten is the *roundest, friendliest* number — then every civilisation should love it, with maybe one or two odd exceptions, at random.

If a *king decreed it* or *everyone copied one village* — then counting should spread like a rumour, from one place outward, along the trade routes.

Then Nia reached the crooked card, and went still.

"If it's fingers," she said slowly, "then people who counted on their bodies *differently...* should have landed on different numbers."

That sentence is a trap — the good kind. It means the FINGERS card can be *tested*. Hunted. Proven wrong, or proven right.

Draw the arrow for your own suspect: *if I'm right, the world should look like...*

Done? Good.

"Open the files," said Sam.

Art: A world map at night, and across every continent, counting systems glowing in three colours — fives here, tens there, twenties scattered like constellations. The pattern is visible from space.

The files are real. Anthropologists — scientists who study how humans live — have recorded the counting systems of hundreds of cultures, on every continent, including peoples who never met anyone outside their own valley.

Here is what the files say.

The counting systems of the world do not land just anywhere. They cluster. Again and again and again, humans have counted in **fives**. In **tens**. In **twenties**.

Sam read the numbers twice. When he spoke, it was almost a whisper.

"Five. Ten. Twenty," he said. "One hand. Two hands. Hands *and feet*."

Cultures separated by oceans, by ice ages, by ten thousand years — all reaching for the same three numbers. The same three numbers every barefoot human on Earth carries around attached to their own body.

Nia felt the back of her neck prickle.

"But—" Sam was still in the files, frowning. "Two of them don't fit."

Art: Two enormous hands fill the page. On the first, the GAPS between the fingers glow — eight valleys. On the second, a thumb touches the segments of the four fingers one by one — twelve little bones — while a distant second hand keeps a tally of five.

File one. The Yuki people of California counted in *eights*.

Eights! No civilisation has eight fingers. The fingers suspect should be finished — until you read *how* the Yuki counted. Not on their fingers.

In the *gaps between* their fingers.

Count your own gaps. Go on — the spaces where a marble would rest. Four on one hand. Eight on two.

File two. Babylon, and its mighty sixty. Watch closely: use your thumb to touch each segment of the four fingers on the same hand — every finger has three little bones. Count them.

Twelve. Now hold up the other hand and keep tally: twelve, once for each of five fingers...

Twelve times five. *Sixty*.

Nia stared at her own two hands as if they had been keeping a secret for eleven years.

Even the exceptions — *especially* the exceptions — were hands.

Go on. You can see it now. Say it out loud —

Art: A cutaway diagram, drawn like an ancient machine. Bedrock below, stamped GROUPING — REQUIRED. Mounted on top, a great brass dial labelled GROUP SIZE — CHOOSE ONE, its rim marked $2 \cdot 5 \cdot 8 \cdot 12 \cdot 20 \cdot 60$, its pointer resting on 10.

Hold on. Before the answer — get the shape of it right, because the shape is the real treasure.

There are two layers to counting, and they are not the same kind of thing.

The bottom layer is *bagging itself* — grouping things up, writing bags first and loose ones after. That layer is bedrock. No civilisation escapes it, because no list of symbols can go on forever. Pentaria needed it. Babylon needed it. You need it.

The top layer is the *size of the bag*. And that layer — as you proved yourself, at a market stall on a world that doesn't exist — is a dial. Mathematics permits every setting. Two works. Five works. Sixty works, with room to spare.

"The game is required," said Nia slowly. "But the team size... the team size was our pick."

Every human civilisation stood in front of that dial.

So who turned it to ten?

Art: FULL SPREAD. Two vast open hands, palms up, fingers spreading toward the sky — and at each fingertip, a numeral kindling like a small sun: 1, 2, 3... to 0. No borders. No boxes. The pages should feel quiet.

You already know. You've known since you counted the gaps.

We count in tens because we count on our hands.

That's the whole secret. The oldest calculator on Earth was never made of brass or silicon. It was standing at the end of every arm, in every valley, in every century — ten fingers, free of charge, wherever a human went.

Egypt never met China. China never met the Maya. They didn't have to. They all consulted the same machine.

Nia looked at her hands for a long moment.

"Four thousand years of mathematics," she said, "because of *these*."

But wait. One thread is still loose, and it leads somewhere unexpected.

Ten never *felt* like a choice, did it? It felt like a law of the universe. Why?

Art: Two shoes on a doormat. One worn soft, moulded to a foot, labelled TEN. One stiff, brand new, labelled FIVE. It is completely obvious which one anybody would reach for.

Think about how walking in your own shoes feels. It doesn't feel like anything. That's the clue.

You have counted in tens since before you could read. Thousands of days. Millions of numbers. And here is the trick your own brilliant brain plays on you:

anything you have done ten thousand times begins to feel like a law of nature.

Ten wasn't a law. It was a shoe — worn so soft, so long, that it stopped feeling like a shoe at all. Remember base five pinching on page 4? That wasn't five being wrong. That was a new shoe.

"So when I signed YES," said Sam, "I wasn't measuring ten. I was measuring how *used to it* I am."

Write down one thing — one word — that once felt impossible for you and is now automatic. Riding a bike. Reading. Swimming.

That word is your proof. It works on numbers too.

And now the twist this mystery has been saving: ten might not even be the *best* setting on the dial.

Art: Two pies. One, under a proud 12, cut into clean halves, thirds, quarters. The other, under a nervous 10, hacked into fifths, one slice visibly unfair. A dozen eggs look on, smug.

Meet twelve — the number that should probably have won.

Split 12 things among 2 friends: six each. Among 3: four each. Among 4: three each. Clean, every time.

Now split 10. Two friends? Fine, five each. Three friends? Someone's getting cheated. Four? Worse.

Try it with real sweets and real friends. Listen for the arguing. Twelve divides beautifully; ten mostly doesn't. Bakers figured this out centuries ago — that's why eggs and doughnuts come in dozens.

"Twelve splits nicer," said Nia, indignant. "Why don't we switch?"

Sam grinned. "Try getting eight billion people to change shoes at the same time."

That's the second secret of the dial. Once a whole world settles on a setting — builds its schools and shops and songs on it — *better* stops mattering. The cost of switching crushes it. Winners aren't always the best. Sometimes they're only the first to get comfortable.

And people really did fight about this. For centuries. Some are still fighting.

Art: Two big wooden bins, freshly painted: NECESSITY and CONVENTION. Nia and Sam are sorting glowing statements into them like post. A few statements hover, undecided.

Time to name what you've captured — two words that most adults mix up their whole lives.

Things that *must* be true, that no civilisation could escape — bagging, carrying, the whole machine — those are **necessity**.

Things that were *chosen* — settled on, gotten used to, until they felt eternal — the size of the bag, the shape of the digits — those are **convention**. And the size of everyone's bag has a name: it's called a **base**. You've been running base ten.

Pentaria runs base five. Babylon ran base sixty.

Warm up your sorting arm:

"You can't write every number with a fixed set of symbols unless you group things." — which bin?

"The number after nine gets written with two digits." — which bin?

(First: necessity. Second: convention — that's ten's shoe talking.)

Two bins. One question — *did the universe insist, or did we?* — and you now own a lens most people never get.

It's older than paper. And it's about to catch something on your own wrist.

Art: A Babylonian astronomer on a ziggurat at night, tracing stars on a clay tablet — and the tablet dissolves into a modern wristwatch and a protractor. Same numbers, four thousand years apart.

Look at your wrist. Or the wall, or a phone's clock screen.

Sixty seconds in a minute. Sixty minutes in an hour. Three hundred and sixty degrees in a full turn — sixty, six times over.

That's not coincidence. That's *Babylon* — the knuckle-counters from page 11 — talking to you across four thousand years. Their empire fell. Their language died. Their cities went under the sand.

Their base sixty is on your wrist right now, keeping perfect time.

"My watch," said Nia, holding it up as if it had introduced itself, "has been speaking Babylonian my whole life."

Conventions are like that. The convenient ones outlive their inventors, hiding inside daily life, pretending to be laws of nature — until a detective with the right lens looks twice.

Today's hunt: catch all three Babylonian numbers — 60, 60, 360 — loose in your own day.

Oh — and that wrong suspect you inked on page eight?

It had famous company.

Art: The corkboard one last time. Beside each old suspect card, a portrait now hangs: robed scholars, wigged philosophers, ink-stained mathematicians — each one holding the same guess in fancier handwriting.

Serious scholars argued for centuries that ten must be nature's favourite — the roundest, most perfect number. Suspect one, in beautiful Latin.

Others wrote that counting spread outward from one gifted civilisation. Suspect three, with footnotes.

Brilliant people. Wrong suspects. The files that settled it — the world's counting systems, gathered valley by valley — took explorers and anthropologists centuries to collect. You crossed in twenty pages a bridge that humanity built over four thousand years.

"So my wrong guess," said Sam, delighted, "had better handwriting in 1650."

That's how this actually works — and it's the part school forgets to mention. Wrong guesses aren't the opposite of discovering. They're the *road*. The only people who never pinned a wrong suspect are the people who never opened the case.

Enough history.

Your new lens needs testing — and Nia had turned the corkboard into a game show.

Art: The bedroom transformed into a glittering quiz set: NECESSITY OR CONVENTION? in lights. Sam at a cardboard podium. Nia as the host, wielding a hairbrush microphone with total authority.

"Welcome," announced Nia, "to the only game show where the buzzer is *thinking*."

Five statements. Two bins. No mercy. Play against Sam — answers in the back, but commit first. Ink means you mean it.

One. *Even numbers always end in 0, 2, 4, 6 or 8.*

Two. *If you multiply any number by its base, its digits all shift one place left.*

Three. *A minute has sixty seconds.*

Four. *"Round numbers" end in zero.*

Five. *No matter the base, adding can make a bag snap shut.*

Sam fell for number one — almost everyone does. It *feels* like bedrock. But it's ten's shoe again: in Pentaria, our six is written "11". Even numbers ending in 0-2-4-6-8 is our choice wearing a law's uniform.

"Round two," said Nia, spinning the board, "raids an old treasure chest."

Art: The bags of beans from The Mystery of Ten, back by popular demand — now wearing tiny detective hats and looking extremely pleased about it.

Some tools from old adventures come when called.

Challenge one. A friend leans over and says: *"Ten is mathematically special. That's why everyone uses it."* You get two sentences and one piece of evidence. Go — out loud or on paper.

(A detective-grade answer sounds something like: *nothing in maths breaks in any other base — I've run base five myself. Ten won because humans carry ten fingers, and the world's counting systems cluster at five, ten and twenty to prove it.*)

Challenge two. Bags and loose, quick-fire: thirteen beans in bags of five — write it the Pentarian way. Twenty-eight beans in bags of ten — write it yours. Feel that? Same trick. Same bedrock. Different dial.

Sam was inventing a third challenge when the letterbox rattled downstairs.

An envelope. Nia's name. Postage stamps she did not recognise — because they had not been printed on this planet.

Inside was a letter that was not written in ten.

Art: The letter, in full: friendly alien handwriting, a doodled self-portrait of ZIB — cheerful, eight-fingered — and a treasure count reading 34, circled twice. Deep-space postmark.

Dear Nia and Sam,

My name is Zib. I have been reading about your world's mathematics — a strange and wonderful system, though you seem weirdly attached to the number ten.

*I am writing to boast, politely: I own **34** gleeps of treasure.*

How many is that in yours?

Yours across the dark, — Zib

At the bottom, a doodle: Zib waving. With four fingers on each hand.

Eight fingers.

"Thirty-four *their* way," said Sam slowly, reaching for the beans. "How much is that *our* way?"

No hints this time. You have everything you need — you've had it since Pentaria. (The back of the book will confirm your answer, but you won't need it, and you'll know when you're right by the click.)

Then write back to Zib — your age, in *Zib's* counting.

Somewhere in that reply, it hits you: you can now write numbers that nobody in your school can read.

Art: Two kids at the back of a classroom passing a folded note dense with digits. The teacher holds it up to the light, baffled but visibly impressed. A tiny "23?" floats above her head.

It began, as these things do, as a secret code.

Birthdays in base five. Locker numbers in base eight. Nia and Sam passing notes that looked like ordinary arithmetic homework and read like nothing on Earth.

"Our birthdays in base five," said Nia, folding the note. "Uncrackable."

"Almost," said Sam, with the smile of someone who now knew exactly *how* crackable — because a code you can build is a code you can break.

Build your own. Pick a base — five, eight, twelve, your choice of dial. Write your age in it. Write the day and month you were born. Hand it to someone and enjoy their face.

Four thousand years ago, turning that dial took an empire.

You did it between two page turns, for fun.

But hold on — because one machine took this trick further than anyone in history dared...

Art: A vast cathedral-dark space inside a computer. Billions of switches like city lights, each one flipped ON or OFF — and together, glowing across the darkness, they spell the reader's own name on a screen.

Somewhere near you right now — in a pocket, on a desk, behind a wall — there is a machine built on the *smallest possible* setting of the dial.

Not ten symbols. Not five.

Two.

On and off. Yes and no. A switch — the only alphabet a machine can feel. Base two: bags of two, bags of bags of two, all the way up.

And with those two symbols, it holds every photograph you've ever taken. Every song you've ever heard. Every game, every map, every film, every message — this entire book, if you're reading it on a screen — spelled out in nothing but *on* and *off*, billions of times a second.

"Two symbols," said Sam, staring at his phone as if it had quietly admitted to being a dragon. "That's all it has. And it runs the *world*?"

It does. And exactly *how* it does — how on-and-off learned to remember, to draw, to sing, to beat grandmasters at chess —

— that story is called ***The Machine That Counts to Two.***

It is even stranger than this one.

Art: The great hands-into-digits image from page 13, now a small poster pinned above Nia's desk, next to two hand-painted bins: NECESSITY / CONVENTION. Below it, a detective's badge, and two signatures — with room for a third.

Every mystery leaves something behind.

This one leaves three things.

A sentence: *we count in tens because we count on our hands.* Say it in ten years; everything else will come back with it.

A lens: two bins, one question — *did the universe insist, or did we?* Point it at clocks. At calendars. At anything wearing a law's uniform. It never wears out.

A badge. Read it carefully, because it says the opposite of what school taught you to be proud of:

I signed my guess in ink. I hunted the evidence myself. I proved my own guess wrong —

— and that's how detectives win.

Nia signed it. Sam signed it — right over the spot where he'd crossed out his YES, which he now considered the best thing on the page.

The third line is blank.

It's yours.

Back matter

[Back matter — as specified in the blueprint: answer key with "where your bags most likely leaked" notes for pages 4, 5, 19, 20, 21; two further challenge sets; Loop schedule: this mystery returns in the clock adventure, the angle adventure, and The Machine That Counts to Two.]